

Multi-Site Outage Post-Incident Report — INC-3312

Affected Sites: DC-North, DC-South

Executive Summary

A regional fiber cut on 22 July disrupted connectivity between DC-North and DC-South, triggering cascading effects at both sites due to a shared authentication service that was, contrary to design intent, only deployed at DC-North without a local fallback at DC-South. This report documents both sites' timelines, impact, and root cause separately, since the two sites experienced different failure modes despite sharing a single triggering event.

DC-North Timeline

- 14:02 — Fiber cut occurs on the regional transport ring due to third-party construction work; automated alert fires within 90 seconds on transport monitoring.
- 14:05 — DC-North duty engineer acknowledges the alert and confirms loss of the primary transport path to DC-South.
- 14:08 — Automatic failover to the backup microwave transport link begins; DC-North's own services remain unaffected since authentication is locally hosted.
- 14:11 — Failover completes; DC-North reports normal operation throughout, aside from the brief 3-minute failover window during which cross-site replication was paused.
- 14:20 — DC-North engineering confirms no customer-facing impact at their site.

DC-South Timeline

- 14:02 — Same fiber cut occurs; DC-South's authentication requests, which depend on the DC-North-hosted authentication service, begin timing out.
- 14:04 — Customer login failures begin appearing across DC-South-hosted applications; monitoring shows failure rate climbing from 0% to 40% within two minutes.
- 14:09 — DC-South on-call escalates to the authentication platform team, who were not initially aware of the fiber cut since their alerts are scoped to application-layer errors, not transport-layer events.
- 14:17 — Platform team identifies that DC-South has no local authentication fallback and begins manually redirecting authentication traffic over the (now-recovered) microwave backup link shared with DC-North's transport recovery.
- 14:34 — Authentication traffic successfully rerouted; login failure rate drops to under 2%.
- 14:41 — Failure rate confirmed at 0%; DC-South incident declared resolved. Total customer-facing impact window at DC-South: from first failure to resolution.

Impact Summary Table

Site	Customer-Facing Impact	Peak Failure Rate	Root Contributing Factor
DC-North	None (internal failover only)	0%	N/A — transport redundancy worked as designed
DC-South	Login failures	40%	No local authentication fallback

Root Cause Analysis

The triggering event — a third-party fiber cut — was outside the organization's control and is not considered part of the root cause. The actual root cause is an architectural gap: the authentication service was designed to be deployed redundantly at both sites, per the original 2023 design document, but the DC-South deployment was deprioritized during the original rollout and never completed. This gap had not caused a visible incident before because DC-North and DC-South transport connectivity had not previously failed since the microwave backup link was commissioned two years ago. The delay between the fiber cut and traffic rerouting (14:02 to 14:34 at DC-South) was extended by the platform team's initial lack of awareness of the transport-layer event, since application monitoring and transport monitoring are currently on separate alerting systems with no cross-linking.

Follow-up Actions

(1) Complete the deferred DC-South local authentication deployment, closing the original 2023 design gap — assigned to the platform team, 6-week deadline. (2) Cross-link transport-layer alerts to application-layer on-call rotations so a transport event automatically notifies affected application teams — assigned to the monitoring team, 3-week deadline. (3) Conduct a full redundancy audit of all shared services between DC-North and DC-South to identify any other single-site dependencies — assigned to the architecture team, 8-week deadline.